

PROFESSIONAL SUMMARY

Master’s student in Mechanical Engineering at Linköping University specializing in Applied Mechanics with a focus on Computational Fluid Dynamics (CFD) and Finite Element Analysis (FEM). Experienced in turbulent flow modeling, heat transfer, thermo-mechanical analysis, and numerical simulation using ANSYS Fluent, ANSYS Mechanical, OpenFOAM, and MATLAB. Possesses industrial experience as a Mechanical Engineer in the oil and gas sector, contributing to engineering design, risk assessment, technical documentation, and project execution.

EDUCATION

Degree / Certificate	Institute / University	Year
M.Sc. Mechanical Engineering	Linköping University (LiU)	Expected 2027
B.Tech. Mechanical Engineering	Indian Institute of Technology (IIT) Indore	2021

Relevant Coursework: Computational Fluid Dynamics (CFD), Advanced CFD, Computational Aerodynamics, Advanced FEM, Continuum Mechanics, Computational Heat Transfer, Road Vehicle Dynamics

WORK EXPERIENCE

Mechanical Engineer – II

iFluids Engineering

Mar 2024 – Feb 2025

- Safety and Risk Assessment:** Developed **Safety Critical Element (SCE)** performance standards using **Bow-Tie analysis** and contributed to engineering risk and performance evaluations.
- Mechanical Design and Documentation:** Performed CAD work and mechanical modelling using **SolidWorks, AutoCAD and CAESAR II**, producing structured technical documentation for performance-critical systems.

Quality Analyst – SME (Mechanical Engineering)

Straive SPI Global

Oct 2021 – May 2022

- Acted as a **Subject Matter Expert** ensuring technical accuracy, clarity and documentation quality.
- Reviewed and validated **500+ mechanical engineering solutions**, strengthening analytical rigour and structured problem-solving.
- Developed quality assurance strategies to improve content accuracy and compliance with engineering standards.
- Collaborated with internal teams to improve workflow efficiency and technical reliability.

ACADEMIC PROJECTS

Fluid-Thermal Analysis of Green Revolution Energy Converter (GREC)

Advanced Project Course | CFD, Heat Transfer & Energy Systems

Aug 2026 – Present

- Developing simulation-based methodologies for thermal-fluid performance evaluation of the **Green Revolution Energy Converter (GREC)**.
- Investigating heat transfer mechanisms, power density and system efficiency through CFD-based modelling, verification and engineering analysis.

Large Eddy Simulation of Periodic Hill Flow

LES CFD | ANSYS Fluent

April 2026 – June 2026

- Developed and validated a **Large Eddy Simulation (LES)** model of turbulent flow over periodic hills.
- Conducted mesh-resolution studies, statistical convergence assessment and validation against DNS reference data.

Vortex-Induced Vibration and POD Analysis

URANS CFD | ANSYS Fluent & MATLAB

Apr 2026 – May 2026

- Developed a 2D transient **URANS CFD model** in ANSYS Fluent to investigate vortex shedding and VIV around a circular cylinder at $Re_D = 10,000$.
- Applied **Proper Orthogonal Decomposition (POD)** using MATLAB and ParaView to identify energy-dominant flow structures.

Transient Thermo-Mechanical Analysis of Gas Turbine Disc

Advanced FEM | ANSYS Mechanical

Jan 2026 – March 2026

- Performed analytical and FEM-based stress analysis of a rotating gas turbine disc under centrifugal and thermal loading.
- Modelled transient temperature fields and stress evolution from cold start to steady operation, including mesh-independence assessment.

Simulation of Aerodynamic and Compressible Flows

Computational Aerodynamics | XFOIL & ANSYS Fluent

Apr 2026 – May 2026

- Evaluated low-speed aerodynamic flows around a NACA airfoil by comparing panel methods in XFOIL with viscous and inviscid CFD approaches in ANSYS Fluent.
- Implemented a finite-difference solver for one-dimensional compressible flow through a de Laval nozzle and analyzed shock-wave formation using artificial viscosity.

TECHNICAL SKILLS

CFD: ANSYS Fluent (RANS,URANS,LES models), ANSYS CFX, OpenFOAM, XFOIL
FEM & Structural Analysis: ANSYS Mechanical (Stress Analysis, Thermal Stress, Transient Analysis, Structural Mechanics, Mesh Independence, Design Verification)
Engineering Software & Programming Languages: ParaView, ANSA, Python, MATLAB, C++ , COMSOL Multiphysics, SolidWorks, AutoCAD, CAESAR II ,